1. Objective:

Understand the mode of operation of counting circuits and decoder.

2. Preparation:

2.1. Read and understand the notes below.

2.2. The schematic of tasks 3.1 – 3.4 can and should be simulated using LTSpice using the files provided for lab 2.

2.3. Set up schematics for all tasks below.
These have to be presented to the tutor before starting the lab exercise.

3. Tasks:

3.1. Asynchronous 4-Bit Binary Down Counter:

Set up an asynchronous 4-bit binary counter on the board using JK flip-flops. Use them as T flip-flops.
After power-on, the counter shall have a defined state (see lab 2).
Use the oscilloscope to visualize the count status (first three bit) and the input.
Use the waveform generator as input. 1 MHz Frequency, the other settings should be obvious.
Which is the propagation delay of the circuit? Which is thereby the maximum frequency? Verify it.

3.2. Asynchronous 4-Bit Binary Down Counter with push-button:

Use a push-button as input. Take care of debouncing it (see notes below).
Use the board's LEDs to visualize the count status.

3.3. Asynchronous 4-Bit Binary Up Counter:

Modify the previous circuit to get an asynchronous 4-bit binary **up** counter.
Check its behavior with the push-button and the LEDs.

3.4. Asynchronous BCD Counter:

Augment the previous circuit to get an asynchronous **BCD** counter.

3.5. Add A Seven-Segment Display:

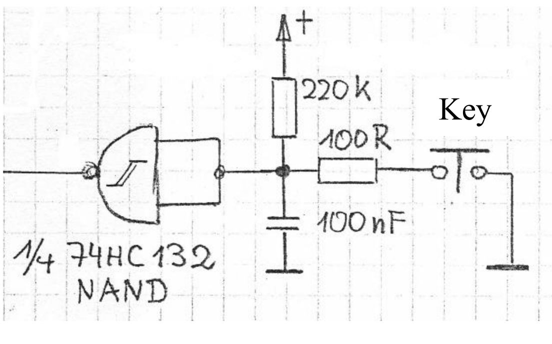
Augment the previous circuit to display the count status using a seven-segment display.
Use the BCD to seven-segment decoder 74LS247.
Argue why the circuit shall or shall not work properly.

4. Report:

Not requested.

5. Notes:

Pressing a push-button causes the voltage to "bounce" almost from one level to another. This can cause a malfunction of the following circuit, since it could be interpreted as being more than one edge.
For this purpose, a **debouncing** circuit is needed.
The circuit shown below eliminates the effect by adding time constants for both ON and OFF. These time constants cause rather slow edges making a Schmitt-trigger input mandatory.



The following integrated circuits can be used:

- 2x 74HC132 (4x 2In NAND gate Schmitt-Trigger)
- 2x 74HC107 (2x J-K-Type flip-flop)
- 1x 74LS247 (1x BCD to 7-segment decoder)
- 1x HDSP-F201 (7-segment LED-display with common anode)